

Kaibalya Sahoo

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PROFESSIONAL SUMMARY

Software Engineer with a strong foundation in data structures, algorithms, and object-oriented design, complemented by hands-on experience building scalable web applications using React.js, Node.js, Nest.js, and PostgreSQL. Proven ability to design and ship secure, production-grade systems — including role-based access control, payment processing, and authentication — across internship and personal-project settings. Comfortable working across the full stack, from problem-solving and system design to frontend UI, backend architecture, database design, caching, and cloud deployment (AWS, Docker). Quick learner focused on writing clean, efficient, well-tested code and collaborating effectively in agile, cross-functional teams.

SKILLS

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Time & Space Complexity Analysis, Problem Solving

Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Responsive UI Design

Backend: Node.js, Express.js, Nest.js, FastAPI, RESTful APIs, WebSockets

Databases: PostgreSQL, MongoDB, QdrantDB (Vector Database), TypeORM

Cloud & DevOps: AWS (EC2, Lambda, ECS), Google Cloud Platform, Docker, Git, GitHub, Vercel, Netlify, Render, CI/CD

Architecture & Tools: Redis, BullMQ, Microservices, Authentication (JWT), Role-Based Access Control, Payment Integration (Razorpay), Webhooks

Generative AI: OpenAI, Gemini, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI

Testing & Practices: Vitest, Unit Testing, Agile/Scrum, Code Review, Technical Documentation

Soft Skills: Problem-Solving, Cross-Functional Collaboration, Communication, Adaptability, Ownership

PROFESSIONAL EXPERIENCE

Software Developer Intern — HyScaler (*Mar 2026 – May 2026*)

- Engineered a role-based access control (RBAC) system supporting 3 permission tiers, reducing unauthorized access incidents and improving admin configuration efficiency.
- Built a payment-splitting and coupon management module using Nest.js and PostgreSQL.
- Implemented JWT-based authentication and authorization, strengthening application security.
- Developed responsive frontend interfaces in React.js with Tailwind CSS, improving page load performance and UI consistency across devices.
- Optimized backend performance by introducing Redis-based queuing and containerized deployments, reducing average API response time by 10%.
- Increased frontend code reliability by writing component tests with Vitest.
- Refactored shared components into reusable, structured modules, cutting duplicate code by 20% and accelerating feature delivery.

PROJECTS

Multi-Vendor E-Commerce Platform — Personal Project — *React.js (Vite), Nest.js, PostgreSQL, TypeORM, Docker, Razorpay*

GitHub: github.com/Sankar110000/Multi-Vendor-Ecommerce-Application

- Designed a full-stack multi-vendor marketplace with 3 distinct roles (Admin, User, Vendor), each with a dedicated dashboard and scoped access, using JWT access and refresh tokens for authentication.
- Built persistent guest cart functionality that merges into the user's account on login, preventing cart loss and improving checkout conversion.
- Developed a vendor module supporting full CRUD on products and coupons, order status updates, and fund withdrawal requests.
- Built an admin module for vendor approval/management and platform-wide payout oversight.
- Integrated Razorpay for customer payments and RazorpayX payouts for vendor withdrawals, with webhook-based verification for both flows.

- Used TypeORM migrations for schema versioning and containerized both frontend and backend with Docker Compose for consistent local and production environments; deployed live on Vercel.

Salon Management System — Personal Project — *React.js, Express.js, MongoDB, JWT*

GitHub: <https://github.com/Sankar110000/salon-management-backend>

- Designed and built a full-featured salon management platform covering appointment booking, staff scheduling, service catalog, and customer records, replacing manual/paper-based scheduling for the target business.
- Implemented a role-based access control system with JWT authentication and authorization, securing access across 2 distinct user roles (admin and staff) and protecting sensitive customer and business data.
- Built a real-time appointment booking flow with conflict detection to prevent double-booking of staff and time slots, improving scheduling reliability.
- Designed normalized MongoDB schemas for bookings, staff, and services to support efficient queries and scale cleanly as the service catalog grew.
- Built a responsive, mobile-friendly React.js interface, enabling salon staff to manage bookings from any device with minimal training.

PDF RAG Chat Application — Personal Project — *Next.js, LangChain, Google Gemini, QdrantDB*

GitHub: github.com/Sankar110000/pdf-rag

- Built a chat-with-your-PDF application that lets users upload a document and ask natural-language questions answered strictly from its contents.
- Built a document ingestion pipeline that parses PDFs with pdf-parse and chunks text using LangChain text splitters for accurate retrieval.
- Generated vector embeddings via Google Gemini and indexed them in QdrantDB, implementing similarity search to retrieve the most relevant chunks per query.
- Built a context-aware answer generation flow that grounds Gemini's responses in retrieved chunks, reducing hallucination compared to a non-RAG baseline.
- Built a Next.js chat interface with Markdown-rendered responses for a readable, ChatGPT-style Q&A experience.

EDUCATION

B.Tech, Computer Science Engineering — Gandhi Institute for Education and Technology, Odisha CGPA: 8.11 | Expected 2026

Higher Secondary (CHSE, XII) — Bidwan Residential College 78.5% | 2022

Secondary (HSE, X) — HRM High School 66.6% | 2020